3/5/2019 First law analysis

- Step thru each component

  inlet/diffuser (0 – 3)

  comb (3 – 4)

  Nozzle (4 – 10)

- We want to allow changes in C_p, but like to retain some simplicity of a CPG.
- Compromise & assume CPG only across components (i.e., C_p is a different <u>const</u> for each component.
- do <u>NOT</u> want to assume isentropic comp./exp → not realistic.
- define component efficiencies η_i such that they are unity or less.

For compression process

$$\eta_c \equiv \frac{\text{isentropic work}}{\text{actual work}} \rightarrow \text{larger.}$$

For expansion process

$$\eta_e \equiv \frac{\text{actual work}}{\text{isentropic work}} \rightarrow \text{larger.}$$

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And in terms of enthalpies,

$$\eta_c \equiv \frac{\overset{\text{isentropic } h_{3s}}{h_{3,s}} - h_0}{\underset{\text{actual } h_3}{h_3} - h_0}$$

~~For~~ And for CPG (componentwise)

$$\eta_c = \frac{T_{3,s} - T_0}{T_3 - T_0} \leq 1$$

$$\eta_e = \frac{h_4 - h_{10}}{h_4 - h_{10,s}} = \frac{T_4 - T_{10}}{T_4 - T_{10,s}} \leq 1$$

<u>Compression Process (0-3)</u>

- External & internal compression
- Interested in p_3/p_0
- Recall $\tau = \frac{T_3}{T_0}$

$$\eta_c = \frac{T_{3,s} - T_0}{T_3 - T_0} = \frac{\frac{T_{3,s}}{T_0} - 1}{\frac{T_3}{T_0} - 1} = \frac{\frac{T_{3,s}}{T_0} - 1}{\tau - 1}$$

But $\left(\frac{T_{3,s}}{T_0}\right) = \left(\frac{p_3}{p_0}\right)^{\frac{\gamma_c - 1}{\gamma_c}}$ Note $\gamma_c = \gamma$ for compression

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Substitute this in & solve

$$\boxed{\frac{p_3}{p_0} = \left\{ \eta_c \left(\psi - 1\right) + 1 \right\}^{\left(\frac{\gamma_c}{\gamma_c - 1}\right)}}$$

Note: Text uses a different def^n of compression efficiency

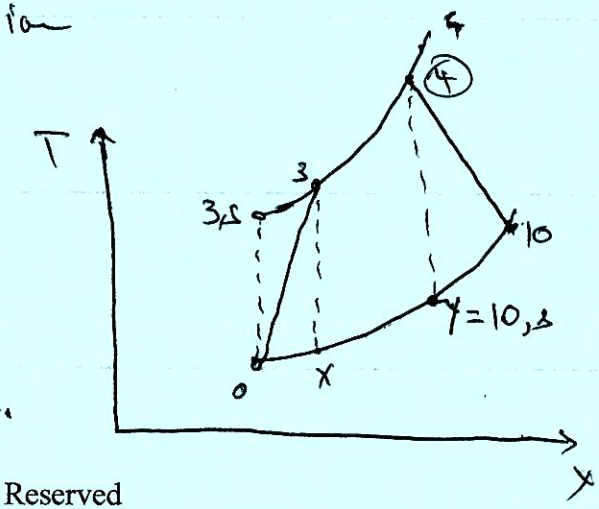
$$\eta_c' \equiv \frac{h_3 - h_x}{h_3 - h_0} \quad \text{where } h_x > h_0$$

$$\eta_e' = \frac{h_4 - h_{10}}{h_4 - h_y} \quad \text{where } h_4 < h_{10}$$

Can show that

$$\left.\begin{aligned} \frac{p_s}{p_0} &= \left(\psi \frac{T_0}{T_x}\right)^{\gamma_c/\gamma_c - 1} \\ \frac{T_x}{T_0} &= \psi\left(1 - \eta_c'\right) + \eta_c' \end{aligned}\right\} \text{Text book version}$$

Look @ T-s diagram to see the difference.



Note $s_x = s_3$

$s_y = s_4 = s_{10,s}$

Can show as η_c & $\eta_c' \to 1$, they are the same.

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In general, they are different b/c

$$h_3 - h_2 \neq h_{3,s} - h_0 \quad \text{(isobars are not parallel in T-s)}$$

$$\eta_c' > \eta_c$$

We use η_c in this course as it is more commonly used in practise.

So for compression

$$\boxed{\frac{p_3}{p_0} = \left\{ \eta_c (\psi - 1) + 1 \right\}^{\frac{\gamma_c}{\gamma_c - 1}}}$$

For combustor $3 \rightarrow 4$

For isobaric heat addition, we have shown that $KE = \text{const}$ $(V_4 = V_3)$

1st law for comb.

$$\dot{Q} - \cancel{\dot{W}}^{\,0} = \dot{m}(\Delta h + \cancel{\Delta KE}^{\,0} + \cancel{\Delta PE}^{\,0})$$

$$\dot{Q} = \eta_b \dot{m}_f H_{vf}$$

What is $\dot{m}$?

$$\dot{m}_3 = \dot{m}_a \ \& \ \dot{m}_4 = \dot{m}_a + \dot{m}_f .$$

If we assume $\dot{m}_f \ll \dot{m}_a$, neglect $\dot{m}_f \Rightarrow \dot{m}_3 = \dot{m}_4 = \dot{m}_a$

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$$\dot{m}_a (h_4 - h_3) = \eta_b \dot{m}_f H_{vf}$$

$$\Rightarrow h_4 - h_3 = \eta_b \underbrace{\left(\frac{\dot{m}_f}{\dot{m}_a}\right)}_{f} H_{vf} = \eta_b f H_{vb}.$$

Due to large changes in temp & composition across comb, let u/s & d/s fluid be CPG, but

$C_{p,c}$ = C_p of comb. inlet fluid

$C_{p,b}$ = C_p of comb. exit fluid.

$$C_{p,b} T_4 - C_{p,c} T_3 = \eta_b f H_{vf}$$

$$\frac{T_4}{T_3} = \frac{C_{p,c}}{C_{p,b}} + \frac{1}{C_{p,b} \cdot T_3} \eta_b f H_{vf} = \frac{C_{p,c}}{C_{p,b}} + \frac{1}{C_{p,b} \cdot \underbrace{\left(\frac{T_3}{T_0}\right)}_{\psi} \cdot T_0} \cdot \eta_b f H_{vf}$$

$$\frac{T_4}{T_3} = \frac{C_{p,c}}{C_{p,b}} + \frac{1}{C_{p,b}} \times \frac{C_{p,c}}{C_{p,c}} \frac{1}{\psi T_0} \eta_b f H_{vf}$$

Note, $C_{p,c}$ is the same as $C_{p,0}$

$$\boxed{\frac{T_4}{T_3} = \frac{C_{p,c}}{C_{p,b}} + \frac{C_{p,c}}{C_{p,b}} \frac{\eta_b f H_{vf}}{\psi T_0 C_{p,0}}}$$

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Recall Gibbs eq^n

$$\Delta s_{3\to 4} = C_p \ln\left(\frac{T_4}{T_3}\right) - \underbrace{R \ln\left(\cancel{\frac{p_4}{p_3}}\right)}_{0} \to 1$$

For simplicity assume $C_{p,b} \approx C_{p,c}$.

$$\Delta s_{3-4} = C_{p,b} \ln\left(\frac{T_4}{T_3}\right)$$

Find Δs for expansion process $(4 \to 10)$

Start w/ $\eta_e \equiv \dfrac{h_4 - h_{10}}{h_4 - h_{10,s}}$

$$= \frac{1 - \dfrac{T_{10}}{T_4}}{1 - \left(\dfrac{T_{10}}{T_4}\right)_s}$$

And $\left(\dfrac{T_{10}}{T_4}\right)_s = \left(\dfrac{p_{10}}{p_4}\right)^{\frac{\gamma_e - 1}{\gamma_e}}$ $\qquad \gamma_e$: γ for expansion process

Solve for $\dfrac{p_{10}}{p_4} = \left\{1 - \eta_e^{-1}\left(1 - \dfrac{T_{10}}{T_4}\right)\right\}^{\frac{\gamma_e}{\gamma_e - 1}}$

don't know what this is!)

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$$\left.\frac{T_{10}}{T_4}\right)_s = \frac{T_{10,s}}{T_4} = \left(\frac{p_{10}}{p_4}\right)^{\frac{\gamma_e - 1}{\gamma_e}}$$

And $\dfrac{p_{10}}{p_4} = \dfrac{p_0}{p_3}$ $\qquad p_{10} = p_0$

$p_3 = p_4$ $\quad$ ($p = \phi$ during comb).

$$\left.\frac{T_{10}}{T_4}\right)_s = \left(\frac{p_0}{p_3}\right)^{\left(\frac{\gamma_e - 1}{\gamma_e}\right)}$$

Subst. for $\left(\dfrac{p_0}{p_3}\right)$

for compression $\qquad$ for exp. process

$$\left.\frac{T_{10}}{T_4}\right)_s = \left\{\eta_c(\tau - 1) + 1\right\}^{-\left(\frac{\gamma_c}{\gamma_c - 1}\right)\left(\frac{\gamma_e - 1}{\gamma_e}\right)}$$

& $$\frac{T_{10}}{T_4} = 1 - \eta_e\left(1 - \left(\frac{T_{10}}{T_4}\right)_s\right)$$

So $$\frac{T_{10}}{T_4} = 1 - \eta_e\left\{1 - \left(\eta_c(\tau - 1) + 1\right)^{-\frac{\gamma_c}{\gamma_c - 1}\cdot\frac{\gamma_e - 1}{\gamma_e}}\right\}$$

Now find Δs.

$$\Delta s_{4-10} = C_{p,e}\ln\left(\frac{T_{10}}{T_4}\right) - R_e \ln\left(\frac{p_{10}}{p_4}\right)$$

$$\Delta s_{4-10} = C_{p,e}\ln\left(\frac{T_{10}}{T_4}\right) - \frac{R_e\gamma_e}{\gamma_e - 1}\ln\left(1 - \eta_e^{-1}\left(1 - \frac{T_{10}}{T_4}\right)\right)$$

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And to complete the cycle, look @ $10 \to 0$

$$\dot{Q} - \cancel{\dot{W}} = \dot{m}\left(\Delta h + \cancel{\Delta KE} + \cancel{\Delta PE}\right)$$

$$q_{10\to 0} = h_{10} - h_0$$

$$= C_{p,r}\left(T_{10} - T_0\right)$$

relaxation

$$= C_{p,r}\, T_0 \left(\frac{T_{10}}{T_0} - 1\right)$$

$$= C_{p,r}\, T_0 \left(\underbrace{\frac{T_3}{T_0}}_{\psi} \cdot \underbrace{\frac{T_{10}}{T_4}}_{f(\eta_i \cdots)} \cdot \underbrace{\frac{T_4}{T_3}}_{f(\eta_i \cdots)} - 1\right)$$

$$\Delta s_{10-0} = C_{p,r} \ln\left(\frac{T_{10}}{T_0}\right) - \underbrace{R \ln\left(\cancel{\frac{p_{10}}{p_0}}\right)}_{0} \to 1$$

$$\Delta s = C_p \ln\left[\psi\left(\frac{T_4}{T_3}\right)\left(\frac{T_{10}}{T_4}\right)\right]$$

For compression $\Delta s = C_{p,c} \ln\left(\underbrace{\frac{T_3}{T_0}}_{\psi}\right) - R \ln\left(\underbrace{\frac{p_3}{p_0}}_{f(\eta_c, \gamma \cdots)}\right)$

$$\Delta s_{tot} = \Delta s_{comp} + \Delta s_{comb.} + \Delta s_{exp.} + \Delta s_{relax}$$

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Thermodynamic cycle efficiency:

$$\eta_{th} = \frac{\text{work out}}{\text{heat in}} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{h_{10} - h_0}{h_4 - h_3} = 1 - \frac{\left(\left(\frac{h_{10}}{h_0}\right) - 1\right) h_0}{\left(\frac{h_4}{h_3} - 1\right) h_3}$$

$h_0 \to C_{p,0} T_0$, $h_3 \to C_{p,c} T_3$

Note $C_{p,c} \approx C_{p,0}$

$$= 1 - \psi^{-1} \left(\frac{\frac{C_{p,e}}{C_{p,0}} \left(\frac{T_{10}}{T_0}\right) - 1}{\frac{C_{p,b}}{C_{p,c}} \left(\frac{T_4}{T_3}\right) - 1} \right)$$

$$\frac{T_{10}}{T_0} = \psi \left(\frac{T_4}{T_3}\right) \left(\frac{T_{10}}{T_4}\right) ;$$

$$\frac{T_4}{T_3} = \frac{C_{p,c}}{C_{p,b}} + \frac{C_{p,c}}{C_{p,b}} \left(\frac{\eta_b f H_{vf}}{\psi C_{p,c} T_0} \right)$$

$$\frac{T_{10}}{T_4} = 1 - \eta_e \left\{ 1 - \left[\eta_c (\psi - 1) + 1 \right]^{-\frac{\gamma_c}{\gamma_c - 1} \cdot \frac{\gamma_e - 1}{\gamma_e}} \right\}$$

$\Rightarrow$ Can see that

$$\eta_{th} = f(\eta_i, C_{p_i}, f, H_{vf}, \psi)$$

$\frac{\eta_b f H_{vf}}{C_{p,c} T_0}$: non-dim. heat input that is typically used.

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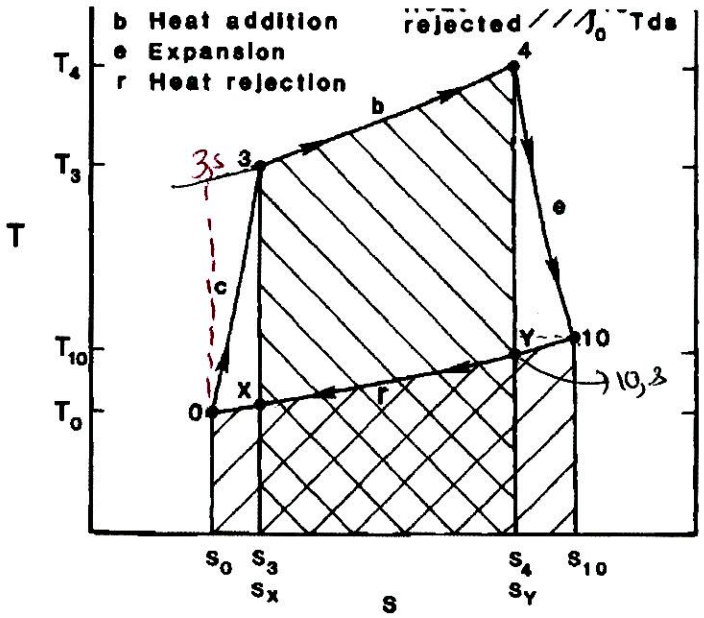

Fig. 4.2 Schematic versions of the airbreathing engine or Bray-